

IOE Entrance Exam Syllabus (BE / B.Arch)

Institute of Engineering, Tribhuvan University - compiled from entrance.ioe.edu.np & official notices (verified 2026-08-22)

Exam Overview

IOE Entrance Exam (BE / B.Arch), Tribhuvan University Institute of Engineering

Mode: Computer-Based Test (CBT) | Duration: 2 hours | Questions: 100 MCQs | Total: 100 marks

Negative marking: 0.25 deducted per wrong answer

Registration: entrance.ioe.edu.np | Fee: Rs. 2,500

Eligibility: 45%+ in +2 with Physics, Chemistry, Mathematics (min C grade), CGPA 2.0+

1. Mathematics (40 questions)

Algebra: sets, complex numbers, polynomials, quadratic equations, sequences & series (AP/GP), binomial theorem, logarithms, matrices & determinants

Trigonometry: identities, inverse functions, heights & distances, general solutions

Analytic Geometry: straight lines, circles, parabolas, ellipses, hyperbolas, 3D coordinates

Calculus: limits, continuity, differentiation (rules, chain, implicit), applications (tangents, maxima/minima), integration (indefinite, definite, areas), differential equations (first order)

Vectors: dot & cross products, scalar triple product, applications

Statistics & Probability: mean/median/mode, dispersion, basic probability

2. Physics (30 questions)

Mechanics: units & dimensions, kinematics, Newton's laws, work-energy-power, rotational motion, gravitation, SHM, elasticity

Heat & Thermodynamics: temperature scales, gas laws, kinetic theory, first & second laws, heat transfer

Waves & Optics: wave motion, sound (Doppler), reflection/refraction, lenses & mirrors, optical instruments, interference & diffraction

Electricity & Magnetism: Coulomb's law, electric fields & potential, capacitance, current & resistance, circuits (Kirchhoff), magnetic fields, electromagnetic induction, AC basics

Modern Physics: photoelectric effect, atomic models (Bohr), nuclear physics, radioactivity, semiconductors (diodes, transistors basics)

3. Chemistry (30 questions)

Physical Chemistry: atomic structure (quantum numbers), periodic table & trends, chemical bonding (ionic/covalent/coordination), states of matter, solutions, chemical equilibrium, electrochemistry, chemical kinetics, thermodynamics basics

Inorganic Chemistry: s/p/d block elements, hydrogen & compounds, metallurgy basics, environmental chemistry

Organic Chemistry: nomenclature (IUPAC), isomerism, alkanes/alkenes/alkynes, aromatic compounds, alcohols/phenols/ethers, aldehydes/ketones, carboxylic acids, amines, polymers, biomolecules (carbohydrates, proteins, fats)

4. English (10 questions — full marks counted in tie-breaks)

Grammar: tenses, subject-verb agreement, articles, prepositions, voice & narration

Vocabulary: synonyms, antonyms, one-word substitution, idioms & phrasal verbs

Reading comprehension & sentence completion

Admission Flow

Entrance exam → merit list (rank) → counselling (multiple rounds) → seat allocation by rank & preference

Tie-break order: higher Mathematics score, then Chemistry, then fewer wrong attempts

Key dates 2083: registration opens Shrawan, exam from Bhadra 11 (examresultnp.com, verified 2026-08-22)